

Nidhi Agrawal

Ph.D Research Scholar, Birla Institute of Technology and Science, Pilani

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🆔 ORCID 📍 Lab- 6021, Disruptive Technologies Lab, BITS Pilani

PROFILE

I am a PhD research scholar under the supervision of Prof. Shashank Gupta at the Disruptive Technologies Lab in the Department of Computer Science and Information Systems at Birla Institute of Technology and Science, Pilani. My broad area of research primarily focuses on Computer Vision (CV), Cyber Physical Systems (CPS), and Vehicle-Bridge Interaction Modelling for long-span bridge assessment.

EDUCATION

Birla Institute of Technology and Science, Pilani <i>Ph.D in Computer Science and Engineering</i>	Rajasthan, India Dec. 2024 – Present
Dr. APJ Abdul Kalam Technical University, Lucknow <i>B.Tech with Hons. in Computer Science and Engineering; CGPA: 8.65</i>	Uttar Pradesh, India Nov 2020 – June 2024
Laxmi Prasad Memorial Public School <i>Intermediate; Percentage: 88.6 (Secured 1st position)</i>	Gorakhpur, India Aug 2018 – June 2019
Don Bosco Sr. Sec. School <i>10th; CGPA: 10</i>	Gorakhpur, India Aug 2016 – June 2017

RESEARCH INTERESTS

Vehicle-bridge Interaction Modeling, Computer Vision in Vehicles, Artificial Intelligence (AI), Machine Learning (ML), Cyber-Physical Systems (CPS).

RESEARCH PROJECTS

Title:	Development of Transfer Learning based Smart Structural Damage Assessment Controller for Autonomous Prediction of Damage Scenarios in Digital Twin-Generated Population of Civil Structures
Funding Agency:	BITS-Cross Disciplinary Research Forum (CDRF)
Sanctioned Amount:	30 Lakhs
PI:	Prof. Shashank Gupta, Department of Computer Science and Information Systems
Co-PI:	Prof. S.N. Patel, Department of Civil Engineering
Co-PI:	Devendra Gokul Patil, Department of Mechanical Engineering
My Role:	Junior Research Fellow (JRF)
Status:	Completed

PUBLICATIONS

- Nidhi Agrawal, Rehan Nath, Aashritha Lakshmi Mallampati, Keshav Goel, Shashank Gupta, Shuvendu Narayan Patel, Devendra Gokul Patil. “*Integrated Finite Element-Based Digital Twin Model and Zero-Shot Transfer Learning Approach for Structural Dynamics and Damage Detection.*” *Journal of Supercomputing*, 2025. (In-press)
- Aditi Singh, Monika Sharma, Nidhi Agrawal, Smriti Sharma, Oshin Rawley, Shashank Gupta, Vinay Chamola. “*Multi-Fidelity Digital Twin-Driven Model for Crack Segmentation and Delamination-Aware Damage Detection*” *Computer & Structures*, 2026. (Under-review)

- Kushi Arun Kumar, Nidhi Agrawal, Ditya Sen, Shashank Gupta. “ *Collision-Severity-Aware Motion Planning Utilising Driving Style Recognition and Intention-Based Trajectory Prediction at Signalised Intersections*” *IEEE transaction on Intelligent Transportation Systems*, 2026.(Communicated)
- Nidhi Agrawal, Abhishek Bajpai, Kumkum Dubey, BDK Patro. "An Effective Approach to Classify Fraud SMS Using Hybrid Machine Learning Models." *IEEE I2CT*, 2023.

WORK EXPERIENCE

Birla Institute of Technology and Science, Pilani.		
Junior Research Fellow (JRF)		Nov. 2024 – March 2025
National Institute of Electronics and Information Technology		
Intern – AI and ML (Information Technology)		Gorakhpur, India Aug. 2024 – Nov 2024

PROFESSIONAL ACTIVITIES

- Presented a paper at International Conference for Convergence in Technology, April 7-9, 2023, in Pune, India, organized by the IEEE Bombay Section.
- Innovation and Incubation Cell Coordinator- Facilitated innovation-driven initiatives, student startups, and entrepreneurship-related programs.
- Cyber Security Club Coordinator- Coordinated technical sessions, awareness programs, and student activities related to cybersecurity practices.

TEACHING ASSISTANTSHIPS

CS F363 - Compiler Construction	Instructor(s): Prof. Shashank Gupta	Spring 2025
CS F351 - Theory of Computation	Instructor(s): Prof. Vishal Gupta	Autumn 2025
CS F363 - Compiler Construction	Instructor(s): Prof. Shashank Gupta	Spring 2026

MENTORING STUDENTS

Keshav, Rehan, Aashritha	Structural Health Assessment utilising Transfer Learning and Finite Element Modeling techniques	(2025)
Saanvi, Deepanpreet	Traffic load Simulation frameworks	(2025)
Arnav	Vehicle-bridge Iteration Modelling	(2026)

TECHNICAL SKILLS

Programming Languages: C, C++, Python, and database management system (Oracle)
Research Software: LaTeX, Mendeley Reference Manager, Smart Draw, Visio
Operating Systems: Basics of MS Windows, Ubuntu
Windows Applications: Microsoft Word, Excel, PowerPoint
Development Tools : Pytorch, Anaconda, Pandas
Web Development : HTML, CSS, Java script

PROFESSIONAL MEMBERSHIPS

- IEEE Student Member
- ACM Student Member

REVIEWER IN PEER-REVIEWED JOURNALS

Engineering Research Express, IOP Publishing
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AWARDS

- 2026: Received 2nd position in Poster Presentation & cash prize of Rs. 2000 at Birla Institute of Technology and Science (BITS) Pilani, Pilani campus, Rajasthan, India.
- 2025: Received Institute Fellowship of Birla Institute of Technology and Science (BITS) Pilani, Pilani campus, Rajasthan, India.
- 2024-25: Received Junior Research Fellowship of Cross Disciplinary Research Forum, Birla Institute of Technology and Science.
- 2022-2023: Institute-Level Group Discussion Competition — Rank Holder & Awardee, REC Kannauj (AKTU), Lucknow.
- 2022-2023: Gold Medalist — Institute-Level Carrom Competition, REC Kannauj (AKTU), Lucknow.

ACADEMIC ACHIEVEMENTS

Secured 1st position in the Intermediate.

Achieved 1st position in the B.Tech Third year.

Achieved B.Tech with Honours in Computer Science & Engineering.

PERSONAL DETAILS

My Name:	Nidhi Agrawal
Gender:	Female
Category:	General
Date of Birth:	11.01.2001
Religion:	Hinduism
Mother's Name:	Aradhana Agrawal
Father's Name:	Vibhuti Agrawal
Nationality:	Indian
Permanent Address:	Ward Number- 19 Gola Bazar, Gorakhpur, UP.
Personal Interests:	Article/Blog writing, Nature enthusiast, Volleyball

REFERENCES

1. **Prof. Shashank Gupta (Supervisor)**, Associate Professor, Department of CSIS, BITS, Pilani.
Email: shashank.gupta@pilani.bits-pilani.ac.in
2. **Prof. S.N. Patel (Co-Supervisor)**, Associate Professor, Department of Civil Engineering, BITS, Pilani.
Email: shuvendu@pilani.bits-pilani.ac.in
3. **Prof. Devendra Gokul Patil (Co-Supervisor)**, Associate Professor, Department of Mechanical Engineering, BITS, Pilani.
Email: devendrap@goa.bits-pilani.ac.in

SELF DECLARATION

I hereby declare that the above information is true to the best of my knowledge and belief.